



UBUNTU DIALOGUE CORPUS

Local + Databricks Parameter Guide

A visual operating manual for moving from raw dialogue to Silver features and Gold summaries.

LOCAL	DATABRICKS	SILVER	GOLD
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ONE PIPELINE. TWO ENVIRONMENTS. FOUR ENTRY POINTS.

Choose your route

The data layers are the same in both environments. What changes is how each stage is launched and how its parameters are entered.

LOCAL

Two PowerShell commands

- 1 Raw CSV -> Silver CSV with `python -m pipeline.pipeline`
- 2 Silver CSV -> one selected Gold CSV with `python -m pipeline.aggregation`
- 3 Use ordinary command-line flags. No JSON brackets.

DATABRICKS

Two dependent job tasks

- 1 Bronze -> Delta Silver with `bronze_to_silver_ubuntu.py`
- 2 Delta Silver -> one selected Delta Gold table with `silver_to_gold_ubuntu.py`
- 3 Use a JSON array of strings in each task's Parameters box.

The order never changes

Silver performs row-level cleaning and enrichment. Gold aggregates that Silver output into one selected analytical grain. Run Silver first, then run Gold once for each output you want.

Local PowerShell

Open PowerShell in the repository root. Flags follow the module name; a backtick continues the same command.

1 BUILD SILVER

```
# Raw CSV -> Silver CSV
& .\venv\python.exe -m pipeline.pipeline
  ".\data\dialogueText_196.csv"
  --sample-size 10000
  --sample-seed 42
  --residual-policy reviewed
  --sentiment vader
  --output-dir ".\outputs\silver_10k"
```

2 BUILD GOLD

```
# Silver CSV -> conversation Gold CSV
& .\venv\python.exe -m pipeline.aggregation
  ".\outputs\silver_10k\silver_ubuntu_dialogue.csv"
  --gold-level conversation
  --output-dir ".\outputs\gold_10k"
```

Using the OpenAI residual-classification option

The pipeline does not open its own key prompt. Run the secure prompt below in the same PowerShell session, place the Silver command inside the try block, and let finally remove the key.

```
$openAiSecureKey = Read-Host "Paste your OpenAI API key" -AsSecureString
$env:OPENAI_API_KEY = [System.Net.NetworkCredential]::new("", $openAiSecureKey).Password
Remove-Variable openAiSecureKey

try {
  # Run Silver here with --residual-policy api.
}
finally {
  Remove-Item Env:OPENAI_API_KEY -ErrorAction SilentlyContinue
}
```

Databricks Jobs

Configure two Python-script tasks. Each task receives its own JSON array in the Parameters textbox.

SETUP

- 1 Upload source data to a Unity Catalog volume and create Bronze, or use a supported CSV/Delta path.
- 2 Add the repository as a Databricks Git folder under Workspace.
- 3 Create a job with a Python script task for Bronze-to-Silver.
- 4 Paste a JSON array of strings into that task's Parameters box.
- 5 Add a dependent Python script task for Silver-to-Gold with its own Parameters JSON.

JSON grammar

Every flag and value is a separate quoted string. Numbers are quoted. Standalone flags have no value after them.

```
[ "--flag", "value", "--standalone-flag" ]
```

OpenAI key

Databricks does not prompt during a job. Never place the key in Parameters JSON. Store it in a Databricks secret scope and expose OPENAI_API_KEY to the Python task.

```
OPENAI_API_KEY={{secrets/<scope-name>/<secret-name>}}
```

Classic compute: configure the secret-backed environment variable on the compute. Serverless: use job-level environment variables when enabled; otherwise use a small secret-reading wrapper.

Remember

Local commands are ordinary CLI syntax. Databricks Python-script parameters are a JSON array of strings. Notebook tasks use a different key-value interface.

Two sampling controls

The same flag name appears at different stages, so read it in context: one changes what reaches Silver; the other creates a sampled Gold output.

BEFORE SILVER

Sample the workload

`--sample-size N --sample-seed S`

Selects N individual input messages before cleaning, enrichment, sentiment, or advanced NLP. Only those messages are written to Silver.

AT GOLD

Create a sample output

`--gold-level sample --sample-size N --random-state S`

Creates a Gold output containing N individual messages selected from Silver. It does not limit any other Gold output.

RAW / BRONZE	SILVER	GOLD
message sample	cleaned + enriched rows	selected analytical grain

Silver reference 1/3

Input, output, execution, and sampling - How data enters, where it lands, and how the work is controlled.

Purpose	Local CLI	Databricks Parameters JSON	Default or requirement
Help	-h / --help	"--help"	Optional
Input CSV	Positional input_csv	"--input-csv-path", "PATH"	Local: required. Databricks: choose one input.
Input table	N/A	"--input-table", "CATALOG.SCHEMA.TABLE"	Choose one input.
Input Delta path	N/A	"--input-delta-path", "PATH"	Choose one input.
Processed-text column	--text-column NAME	N/A	text_cleaned
Source-text column	--source-text-column NAME	N/A	text
Local output folder	--output-dir PATH	N/A	outputs
Output table	N/A	"--output-table", "CATALOG.SCHEMA.TABLE"	Choose one output.
Output Delta path	N/A	"--output-delta-path", "PATH"	Choose one output.
Audit table	N/A	"--audit-table", "TABLE"	Required in Databricks; automatic locally.
Metrics table	N/A	"--run-metrics-table", "TABLE"	Optional
Write mode	N/A	"--write-mode", "errorifexists overwrite append"	errorifexists
Local parallel mode	--parallel auto on off	N/A	auto
Local process count	--workers N	N/A	Automatic, conservatively capped
Spark partitions	N/A	"--repartition-count", "N"	Optional positive integer
Input-message sample	--sample-size N	"--sample-size", "N"	All messages
Input sample seed	--sample-seed S	"--sample-seed", "S"	0
Intermediate strategy	N/A	"--materialization-mode", "auto persist delta none"	auto
Scratch-table schema	N/A	"--materialization-schema", "CATALOG.SCHEMA"	workspace.default

Silver reference 2/3

Residual classification - Human review, reviewed overlays, and optional OpenAI classification.

Purpose	Local CLI	Databricks Parameters JSON	Default
Policy	--residual-policy manual_review reviewed api	"--residual-policy", "VALUE"	manual_review
Maximum NONWORD rate	--maximum-nonword-token-rate X	"--maximum-nonword-token-rate", "X"	0.25
API model	--residual-api-model NAME	"--residual-api-model", "NAME"	gpt-4o-mini-2024-07-18
API batch size	--residual-api-batch-size N	"--residual-api-batch-size", "N"	500 unique terms
API timeout	--residual-api-timeout-seconds X	"--residual-api-timeout-seconds", "X"	120.0 seconds
API retries	--residual-api-max-retries N	"--residual-api-max-retries", "N"	2
Residual vocabulary ceiling	N/A	"--maximum-residual-vocabulary", "N"	500000
API key	Environment variable OPENAI_API_KEY	Secret-backed OPENAI_API_KEY	Never a CLI/JSON parameter

Silver reference 3/3

Sentiment transformer - VADER/transformer selection and reproducible transformer settings.

Purpose	Local CLI	Databricks Parameters JSON	Default
Sentiment mode	--sentiment none vader transformer both	"--sentiment", "VALUE"	vader
Model	--transformer-model NAME	"--transformer-model", "NAME"	cardiffnlp/twitter-roberta-base-sentiment-latest
Revision	--transformer-revision REV	"--transformer-revision", "REV"	Pinned commit 3216a57f...57c20c49fb4bf9c7
Batch size	--transformer-batch-size N	"--transformer-batch-size", "N"	32
Inference chunk	--transformer-inference-chunk-size N	"--transformer-inference-chunk-size", "N"	4096
Device	--transformer-device N	"--transformer-device", "N"	Automatic; -1 CPU, 0+ GPU index
Dtype	--transformer-dtype float32 float16 bfloat16	"--transformer-dtype", "VALUE"	float32

Silver reference: advanced NLP

Optional spaCy, topic-modeling, and transformer-emotion controls. Leave these disabled for the lightest CPU path.

Purpose	Local CLI	Databricks Parameters JSON	Default
Enable advanced NLP	--advanced-nlp	"--advanced-nlp"	Off
Disable spaCy	N/A	"--no-spacy"	Off; local CLI runs spaCy with advanced NLP.
spaCy model	--spacy-model NAME	"--spacy-model", "NAME"	en_core_web_sm
spaCy batch size	N/A	"--spacy-batch-size", "N"	128
Include token details	N/A	"--include-token-details"	Off
Topic mode	--topics none nmf	"--topics", "VALUE"	none
Topic-fit sample	N/A	"--topic-fit-sample-size", "N"	100000; affects NMF fitting, not Silver row count.
Topic batch size	N/A	"--topic-batch-size", "N"	50000
Minimum topic coverage	--topic-minimum-vocabulary-coverage X	"--topic-minimum-vocabulary-coverage", "X"	0.50
Emotion mode	--emotion none transformer	"--emotion", "VALUE"	none
Emotion model	--emotion-model NAME	"--emotion-model", "NAME"	j-hartmann/emotion-english-distilroberta-base
Emotion revision	--emotion-revision REV	"--emotion-revision", "REV"	Pinned commit cea2f78...e00ef93811f6eb
Emotion batch size	--emotion-batch-size N	"--emotion-batch-size", "N"	32
Emotion inference chunk	--emotion-inference-chunk-size N	"--emotion-inference-chunk-size", "N"	4096
Emotion device	--emotion-device N	"--emotion-device", "N"	Automatic; -1 CPU, 0+ GPU index
Emotion dtype	--emotion-dtype float32 float16 bfloat16	"--emotion-dtype", "VALUE"	float32

Programmatic-only local controls

PipelineConfig also supports spacy_batch_size, include_token_details, topic_fit_sample_size, topic_batch_size, and validate_advanced_nlp. The local CLI does not expose them.

Gold parameter reference

Run the Gold entry point once for each analytical output you want. Local and Databricks expose the same grains with environment-specific I/O.

Purpose	Local Gold CLI	Databricks Parameters JSON	Default or requirement
Help	-h / --help	"--help"	Optional
Silver CSV	Positional input_csv	N/A	Required locally
Silver table	N/A	"--input-table", "TABLE"	Choose one Databricks input.
Silver Delta path	N/A	"--input-delta-path", "PATH"	Choose one Databricks input.
Local output folder	--output-dir PATH	N/A	outputs
Gold output table	N/A	"--output-table", "TABLE"	Required
Audit table	N/A	"--audit-table", "TABLE"	Databricks default: <output_table>_audit; automatic locally.
Metrics table	N/A	"--run-metrics-table", "TABLE"	Optional
Gold level	--gold-level LEVEL	"--gold-level", "LEVEL"	Required
Date grain	--date-granularity day week month	"--date-granularity", "VALUE"	day; date Gold only
Channel column	--channel-column NAME	"--channel-column", "NAME"	channel; channel Gold only
Release axis	--release-axis since until	"--release-axis", "VALUE"	since; release Gold only
Sample-Gold size	--sample-size N	"--sample-size", "N"	Required only when Gold level is sample.
Sample-Gold seed	--random-state S	"--random-state", "S"	0
Skip local feature engineering	--no-feature-engineering	N/A	Feature engineering runs locally by default.
Intermediate strategy	N/A	"--materialization-mode", "auto persist delta none"	auto
Scratch-table schema	N/A	"--materialization-schema", "CATALOG.SCHEMA"	workspace.default
Write mode	N/A	N/A	Both replace their named Gold output; no user-facing flag.

Choose a Gold grain

Each Gold level produces one row per selected analytical unit.

Level	One output row per...
conversation	distinct conversation represented in Silver
user	user represented in Silver
date	selected day, week, or month
channel	value in the selected channel column
release	Ubuntu release-cycle bucket
language	language/script label

Level	One output row per...
residual	residual term/classification
technical	technical term/category
topic	fitted topic
entity	named entity
sample	selected Silver message

Fast selection rule

Choose conversation, user, date, channel, or release for operational summaries. Choose language, residual, technical, topic, or entity for NLP distributions. Choose sample only when you want a message-level Gold extract.

Six rules to remember

If nothing else stays in memory, keep these six distinctions.

LOCAL	Ordinary PowerShell command. No JSON brackets.
DATABRICKS	JSON array of strings. Quote numbers too.
STANDALONE FLAG	One JSON item: "--advanced-nlp".
SILVER SAMPLE	--sample-seed controls pre-Silver message selection.
GOLD SAMPLE	--random-state controls only --gold-level sample.
OPENAI KEY	Environment variable or Databricks secret. Never CLI, JSON, Git, or audit output.

Verify the installation

Run the complete tracked test suite from the repository root.

```
.\.venv\python.exe -m unittest discover -s tests -v
```

Source of truth

This guide is generated from tools/pdf_sources/local_and_databricks_parameters.md and the current pipeline interfaces. Rebuild it after parameter changes so the PDF and executable entry points remain aligned.